



CSE461

Introduction to Robotics Lab

Real-World Implication

Group: 5

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Submitted to -

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1. Introduction:

Our project is an independent garbage collecting robot that is capable of picking up small pieces of garbage from the ground and loading it into an onboard bucket using a mechanical claw. The robot is constructed around the use of an arduino-uno and incorporates an ultrasonic sensor, an IR obstacle sensor and a load cell to autonomously navigate, distinguish between wall, furniture and a garbage container and detect when the garbage container is full. When it feels that there is a small object in front of it, it lowers one claw, clutches the object and then lifts it up and puts it in the bucket, without anyone directing it. In this report, the various real-world implications of creating and ultimately putting into use such a robot are explored, including the impact on society, health, safety, law and culture if the robot were to be deployed in the real world.

2. Purpose:

This robot is being designed for the purpose of demonstrating that a task that is typically performed manually, namely the collection of small litter from a floor or open area, can be performed by a low cost autonomous system. Garbage collection is currently primarily manual in the public areas of university corridors, parks or footpaths and requires a lot of time and effort and is hazardous to health for workers. Our robot is designed to be a small scale example to show how such a task might be automated with simple sensors and actuators, and without the expensive vision systems that are usually required for such a task. If this system is cleaned and expanded, it could benefit in the number of people required to clean shared areas and may also benefit the consistency of the clean, as the robot doesn't get tired, or miss spots as a person might after doing the same repetitive task for hours.

3. Societal implications:

If such a robot could be seen in lots of locations, such as campus, workplaces or parks, the cleaning process will be organized in a new way. Cleaning teams would not have to spend as much time collecting small pieces of litter and would be able to dedicate more time to tasks that are not possible for a machine, such as deep cleaning, or dealing with larger waste pieces. At the same time, there is a real fear of job loss, as cleaning is one of the most common jobs for many, particularly in a country where manual labour remains a significant component of the workforce such as Bangladesh. Cleaner public areas provided by robots, however, can enhance public areas' overall appearance and cleanliness which can lead to increased comfort in using parks, campuses or walkways.

Our group has seen the advantages of automatic litter collection, but we also understand the need to take into account the impact of such robots when they are rolled out in large numbers, on the people who are currently doing the job.

4. Health implications:

In terms of health, this robot could be quite beneficial as hand men may come in contact with bacteria or sharp objects, or other contaminants in the garbage. A small ball of litter is collected by the claw arm of a robot, eliminating the need for workers to handle it with bare hands, which reduces the risk of getting cuts, infections or exposure to unhygienic waste. But it's not risk free for the robot, either, because the claw and arm operate with servo motors and if someone put their fingers near the claw while it was moving, there is a small chance that it would pinch them. One worry is that if the robot's sensors are deceived by some waste material such as broken glass or something sharp, it may fail to deal with it appropriately and harm itself or spread it even more. Overall, a reduction of direct human contact with unsanitary waste is achieved, but the robot does cause a few minor mechanical dangers that would have to be incorporated in later versions through better sensor calibration.

5. Safety implications:

This was one of the larger things that we had to consider when designing this robot, especially since it is operating autonomously and is located in shared areas and touching people. The ultrasonic sensor at the front is intended to prevent the robot from colliding with people, walls or larger obstacles, and the IR sensor on the bottom is only for detecting small items on the ground, garbage. Nevertheless, the robot could still get the wrong object if, for example, a person walks near the robot when it is scanning the ground, and the claw then catches a foot or shoe instead of the garbage. The overall safety concern of a moving robot in a crowded corridor is also present; if the ultrasonic sensor was unable to detect the presence of a moving obstacle in time, collision with it may still occur. For this reason, our team believes any implementation of this robot outside of a lab setting would require additional safety features including reduced speed in crowded location, an improved range of obstacle detection, and possibly a manual override or emergency stop button to ensure that the robot can be manually controlled at all times in case of an emergency.

6. Legal implications:

This robot is just a prototype, designed to demonstrate the robot in the lab, so it is not in the same legal category as a robot designed for use in the real world in a public place, but there are some questions to consider if the robot is used as a real-world public space robot. Just as other autonomous vehicles are tested for safety before entering public areas, any robot navigating human populated environments such as campus, parks or footpaths would probably have to pass a safety certification for it to be safe to operate in human populated areas and to not harm people or property. Also, there is the question of liability, and who would be responsible if the robot is designed to crash in an accident, or to damage something if it runs independently, or is responsible for the robot. In this case, privacy is not really a concern as our robot does not use cameras and doesn't record any personal information, but if a future version of our robot added a camera for improved object detection, then privacy would also be relevant here. These legal implications are not so much a concern at this moment, since the robot will be tested only in the lab and controlled settings, but rather a future consideration.

7. Cultural implications:

As cleaning tasks in Bangladesh and other societies, where there is a strong interweaving between manual labor and cultural and social norms, are likely to be perceived differently with the addition of autonomous cleaning robots, these robots could be useful in changing the attitude of people towards cleanliness jobs. Cleaning work is sometimes considered as low status labour, and if the robots will start doing the cleaning, it will probably change its public image, in the long term, since the job itself will not be the one occupied by certain people but rather by robots. Then again, there might be some opposition from communities that perceive robots taking the place of humans they know, particularly within smaller communities where work in the cleaning industry is scarce and low-paid, offering a reliable source of income for numerous families.

8. Conclusion:

Overall, our autonomous garbage collecting robot with a claw is a low-cost, practical solution to automate a small segment of waste management and, if improved, could provide significant advantages in terms of hygiene, efficiency and worker safety. Concurrently, our group knows that any technology that is intended for actual public application has implications that extend beyond the technical ones, such as job impacts, safety issues around humans, legal issues that arise, and social/cultural reactions to the technology. It was a good thought exercise to consider not just a working robot for the lab

but what would have to be done if such a project were implemented at a larger scale and beyond the confines of the classroom.